

Q-1 What is Relation and Domain, explain in details.

Q-2, Find out the domain & Range

(i) $\{(a,1) (b,2) (c,3)\}$

(iii) $\{(x,11) (y,100) (z,188)\}$

(ii) $\{(p,10) (q,12) (r,18)\}$

(iv) $\{(M,88) (N,98) (O,99)\}$

Q-3 Explain the all types of Relations.

Q-4 Explain the Binary Relation in a set

(i) Reflexive

(iii) Transitive

(ii) Symmetric

Q-5 gives the value, create a matrix and graph

(i) $(1,2) (1,5) (1,4) (2,4) (2,3) (3,3) (3,4) (5,1) (5,2)$
 $(4,1) (4,2) (4,5)$

(ii) $(2,1) (2,2) (1,1) (1,2)$

Q-6 $X = \{1, 2, \dots, 7\}$

$R = \{ \langle x, y \rangle \mid x-y \text{ is divisible by } 3 \}$
 draw graph

Q-7 Draw Hasse diagram

Let, (i) $P = \{1, 2, 3, 4\}$

(ii) $X = \{\emptyset, a, b, (a,b), (a,b,c)\}$

(iii) $A = \{1, 2, 3, 6\}$

(iv) $D = \{ a,b,c, (a,b) (b,c)$
 $(a,c) (a,b,c)$
 $(b,c,a) \}$

(1) ~~find~~

Q-8. Find out the GLB and LUB.

① $A = \{1, 2, 3, 5, 30\}$

② $A = \{4, 6, 5\}$

③ $A = \{1, 2, 3, 5, 15, 6, 30\}$

Q-9 MATRIX multiple

① $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

② $R_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, R_p = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

